

Global Warming Science

<https://courses.seas.harvard.edu/climate/eli/Courses/EPS101/>

Ice sheets!

Please use the template below to answer the workshop questions. “XX” indicates places where you need to complete/write code or add a discussion.

your name:

```
[ ]: # import needed libraries and load data:
import numpy as np
import pickle
import matplotlib.pyplot as plt
from matplotlib import ticker, cm, colors
import cartopy.crs as ccrs
import cartopy.feature as cfeature
from cartopy.feature import NaturalEarthFeature
from cartopy import config
from mpl_toolkits.axes_grid1 import make_axes_locatable

# load variables from a pickle file:
with open('./ice_sheets_variables.pickle', 'rb') as file:
    d = pickle.load(file)
    # print information about each extracted variable:
    for key in list(d.keys()):
        print(f"extracting pickled variable: name={key}; size={d[key].shape}")
        #print("; type=", type(d[key]))
    globals().update(d)
```

Explanation of input variables:

AIS=Antarctic ice sheet; GIS=Greenland ice sheet

observed SMB from GRACE gravity satellite: AIS_GRACE_SMB, AIS_GRACE_years

GIS_GRACE_SMB, GIS_GRACE_years

Antarctica SMB time series and map:

AIS_evap_timeseries_annual_avg, AIS_evap_timeseries_annual_std, AIS_net_SMB_timeseries_annual_avg,
AIS_net_SMB_timeseries_annual_std

AIS_net_SMB_map, AIS_net_SMB_map_lat, AIS_net_SMB_map_lon

Greenland SMB time series and map:

GIS_evap_timeseries_annual_avg, GIS_evap_timeseries_annual_std, GIS_net_SMB_timeseries_annual_avg,
GIS_net_SMB_timeseries_annual_std

GIS_net_SMB_map, GIS_net_SMB_map_lat, GIS_net_SMB_map_lon

Surface air temperature data as a function of day of year, lat, lon and the corresponding axes:

SAT_control, SAT_rcp85

SAT_lat, SAT_lon, SAT_time

SMB time axis: SMB_timeseries_annual_years

land masks:

landice_antarctica_mask_control, landice_antarctica_mask_rcp85, landice_greenland_mask_control, landice_greenland_rcp85

lon/lat axes: one over the relevant land ice sheet, zero elsewhere

landice_lat, landice_lon

1) Observations and projections:

1a) Plot the observed mass changes in Greenland and Antarctica from the GRACE mission.

```
[ ]: # plot
fig=plt.figure(figsize=(6,2),dpi=300)

plt.subplot(1,2,1)
plt.plot(XX GRACE, 'b', lw=0.5)
axes=plt.gca()
plt.xlabel(XX)
plt.title("(a) Greenland", loc="left")
plt.gca().set_xticks(np.arange(2001,2020,1), minor=True)
plt.xlim(XX,XX)
plt.ylabel("Mass (Gt)")

plt.subplot(1,2,2)
plt.plot(XX AIS_GRACE, 'b', lw=0.5)
axes=plt.gca()
plt.xlabel(XX)
plt.title("(b) Antarctica", loc="left")
plt.xlim(XX,XX)
```

1b) Contour the projected net SMB change under the RCP8.5 scenario from 1920 to 2100, for Greenland and for Antarctica. Plot time series of total net SMB over Greenland and over Antarctica for the same scenario. Explain your plots, discussing and rationalizing both the spatial and temporal structure of the projected changes.

```
[ ]: # Contour SMB change maps and plot time series of SMB changes:
# -----

# initialize figure:
fig = plt.figure(figsize=(7,5.5),dpi=300)

# Antarctica contour:
# -----
ax = plt.subplot(2,2,1,projection=ccrs.SouthPolarStereo(central_longitude=0.0 \
                                                         , globe=None))
ax.set_extent([0, 359.99, -90, -60], crs=ccrs.PlateCarree())
ax.coastlines(resolution='50m',lw=0.25,color="gray")
plt.set_cmap('bwr_r')
#plt.set_cmap('Blues')
ax.gridlines(linewidth=0.2)
DATA1=
levels=XX
c=ax.contourf(XX AIS_net_SMB_map \
              ,transform=ccrs.PlateCarree(),extend='both' \
```

```

,levels=levels)

# Greenland contour:
# -----
# plot:
ax2 = plt.subplot(2,2,2,projection=ccrs.PlateCarree())
ax2.set_extent([284, 350, 57, 85], crs=ccrs.PlateCarree())
ax2.coastlines(resolution='50m',lw=0.25,color="gray")
DATA1=
levels=XX
c=ax2.contourf(XX GIS_net_SMB_map, transform=ccrs.PlateCarree(),extend='both'\
,levels=levels)
clb=plt.colorbar(c, shrink=0.82, pad=0.05, ax=ax2, ticks=levels, format='%.2g')

# Antarctica timeseries:
# -----
# plot:
ax = plt.subplot(2,2,3)
# net SMB:
plt.plot(XX AIS_net_SMB_timeseries_annual_avg)
# evaporation/sublimation:
y=AIS_evap_timeseries_annual_avg
plt.plot(XX SMB_timeseries_annual_years)
plt.xlabel(XX)
plt.ylabel("SMB (Gt/year)")
plt.xlim(XX,XX)

# Greenland timeseries:
# -----
# plot:
ax = plt.subplot(2,2,4)
# net SMB:
plt.plot(XX GIS_net_SMB_timeseries_annual_avg)
# evaporation/sublimation:
plt.plot(XX GIS_evap_timeseries_annual_avg)
plt.xlabel(XX)
plt.xlim(XX,XX)

```

2) Ice stream acceleration:

By what factor should ice streams in Greenland accelerate now in order to contribute a 1.5 m sea level rise by 2100.

1.5 m of sea level rise corresponds to $\rho_{water} 70\% \cdot (4\pi R_e^2) \cdot 1.5$ of mass loss (70% comes from the percentage area on Earth covered by ocean, and R_e is the radius of the Earth). \$\$\$\$

```

[ ]: rho_ice=900
rho_water=1000
Re=6371000 # earth radius
dh=1.5 # change in sea level

```

```

# first, calculate how many Gt ice correspond to the specified sea level rise
ice_mass_loss_equiv_to_dh_Gt=XX/1.0e12
print("mass loss in Gt corresponding to %3.3g meter of sea level rise: %3.3g\
→Gt"%(dh,ice_mass_loss_equiv_to_dh_Gt))

# Currently, the Greenland ice sheet accumulates at a rate of 520 per year,
# and ablates (including runoff, basal melt and iceburg production) at a rate of 720
→per year,
# yielding roughly 200 per year of mass loss:

current_accumulation_rate=520
current_yearly_ablation_Gt=720
# the needed ablation rate per year over the years until 2100 is therefore:
needed_yearly_ablation_rate=XX

acceleration_factor=needed_yearly_ablation_rate/current_yearly_ablation_Gt
print(f"needed acceleration factor is {acceleration_factor:3.3g}")

```

3) Positive Degree Days:

3a) Calculate and plot PDD maps for Greenland and Antarctica for present-day, and for year 2100 based on temperature projection from the RCP 8.5 scenario.

```

[ ]: # calculate PDD maps for control and rcp85:
pdd_map_control=np.sum((SAT_control-273.15)*(SAT_control>273.15),axis=0)
pdd_map_rcp85=XX

# apply masks for greenland and antarctica before plotting, so that ocean areas are
→not contoured too:
pdd_map_greenland_control=pdd_map_control*(1.0*mask_landice_greenland_control)
pdd_map_antarctica_control=XX
pdd_map_greenland_rcp85=XX
pdd_map_antarctica_rcp85=XX

# plot Greenland present-day (control):
# -----
lons, lats = np.meshgrid(lon, lat)
fig = plt.figure(dpi=150)#figsize=(6, 4))
ax = plt.axes(projection=ccrs.PlateCarree(central_longitude=0.0, globe=None))
ax.set_extent([-70, -10, 50, 85], crs=ccrs.PlateCarree())
ax.coastlines(resolution='110m')
ax.gridlines()

cls=10*np.arange(-3,3,0.4)
c=plt.contourf(lons, lats, np.maximum(pdd_map_greenland_control,0.0001),levels=cls\
,transform=ccrs.PlateCarree()
,norm=colors.LogNorm(min(cls),max(cls)), extend='both', cmap='RdBu_r')
plt.xticks(np.arange(-60,0,20))
plt.yticks(np.arange(50,90,10))
plt.title('current PDD (°C day)');
plt.colorbar(c, shrink=0.7, pad=0.02)
plt.pause(0.05)

```

```
# plot Greenland RCP8.5:
# -----
fig = plt.figure(dpi=150)#figsize=(6, 4))
#XX
```

```
[ ]: # plot Antarctica present day (control):
# -----
fig = plt.figure(figsize=(3,3),dpi=200)
ax = plt.axes(projection=ccrs.SouthPolarStereo(central_longitude=0.0, globe=None))
ax.set_extent([0, 359.99999, -90, -60], crs=ccrs.PlateCarree())
ax.coastlines(resolution='110m')
ax.gridlines()
cls=10*np.arange(-3,3,0.4)
c=plt.contourf(lons, lats, np.maximum(pdd_map_antarctica_control,0.0001),levels=cls\
,transform=ccrs.PlateCarree()\
,norm=colors.LogNorm(min(cls),max(cls))\
,extend='both', cmap='RdBu_r')
plt.title('current PDD (°C day)');
cbar = plt.colorbar(c, shrink=1.0, pad=0.02);

# plot Antarctica RCP8.5:
# -----
fig = plt.figure(figsize=(3,3),dpi=200)
#XX
```

3b) Evaluate the expected acceleration of melt rate (assuming it is proportional to the PDD change), and the consequences to sea level rise.

```
[ ]: # estimate total PDD over each ice sheet:

pdd_total_greenland_control=np.sum(pdd_map_greenland_control,axis=(0,1))
pdd_total_greenland_rcp85=XX
pdd_total_antarctica_control=XX
pdd_total_antarctica_rcp85=XX

print(f"current summed pdd on greenland ice sheet: {pdd_total_greenland_control:3.3g}")
print(f"future summed pdd on greenland ice sheet: {pdd_total_greenland_rcp85:3.3g}")
print(
    f"current summed pdd on antarctica ice sheet: "
    f"{pdd_total_antarctica_control:3.3g}"
)
print(f"future summed pdd on antarctica ice sheet: {pdd_total_antarctica_rcp85:3.3g}")
print(
    f"PDD on greenland ice sheet is predicted to increase by a factor of "
    f"{pdd_total_greenland_rcp85 / pdd_total_greenland_control:3.3g}"
)
print(
    f"PDD on antarctica ice sheet is predicted to increase by a factor of "
    f"{pdd_total_antarctica_rcp85 / pdd_total_antarctica_control:3.3g}"
)
```

3c) Compare your SMB results for Greenland and Antarctica to the above calculated required acceleration of ice streams, and speculate which mechanism is likely to contribute a large sea level increase by 2100. XX

4) Calving

4a) Buoyancy forcing: What is the maximum protruding section length L in Fig. 1c that can be stable to buoyancy forces. XX develop the needed formulas here

```
[ ]: tau_c=1e5
rho_ice=900
rho_water=1000
g=9.8
print(f"submerged shelf height={XX:g}")
```

4b) Hydro-fracturing and buoyancy forcing: Consider grounded ice of height $H = 2$ km, bordering with sea water such that the top of the ice is above sea level. Calculate the minimum depth of water D that will lead to the flotation and therefore breakup of the ice, taking into account the adhesion to the rest of the ice via the yield stress τ_c . How would your answer change if there was a hydro-fracture extending to half the thickness of the ice? Let $L = 150$ m be the distance from the crack to the ocean.

ANSWER: XX

```
[ ]: H=2000
L=150
delta=0.5
D=XX
print(f"D={D:g}")
```

4c) Cliff instability: What is the highest cliff height H in Fig. 1b that allows the marked 45° section to be stable. The projected gravity acceleration rate along the slope is $g \sin(45^\circ) = g/\sqrt{2}$, and thus the down-slope force is XX. On the other hand, the maximum tangential stress can be supplied by the underneath ice shell is XX. Thus the section will break if

$$XX > XX$$

or when

$$L > XX$$

```
[ ]: # calculate here:
tau_c=1e5
rho_ice=900
rho_water=1000
g=9.8
print(f"max cliff height={XX:g}")
```

5) Basal hydrology

5a) Basal temperature and melting: Calculate the basal temperature, and where relevant also the melt rate, given a surface ice temperature of $T_s = -40$ C, geothermal heat flux of $G = 0.07$ W/m² and an ice thicknesses of $H = 1, 2, 3$ km.

```
[ ]: kappa_ice=2.2 # W/m/K
G=0.07
Ts=-40
T_1km=XX
```

```

T_2km=XX
T_3km=XX

print(f"Basal temperature is {T_1km:3.2f} C at the base of a 1-km thick ice shell.")
print(f"Basal temperature is {T_2km:3.2f} C at the base of a 2-km thick ice shell.")
print(f"Basal temperature is {T_3km:3.2f} C at the base of a 3-km thick ice shell.")

```

5a solution continued: calculate the basal melt rate The basal temperature of an ice shell of XX km thick is above freezing point and thus melting will occur. The melting rate is determined by the amount of geothermal heat flux that cannot be conducted upward while holding the basal temperature at the freezing point 273.15~K.

$$M/L = XX$$

Finally, M/L needs to be converted to m/year, XX

```

[ ]: # calc melt rate:

L=336000 # J/kg
sectoyear=86400*365
rho_ice=900
M_XXkm=XX # m/year

print(f"Basal melting rate is {M_2km:3.2g} m/year beneath a 2-km thick ice shell.")
print(f"Basal melting rate is {M_3km:3.2g} m/year beneath a 3-km thick ice shell.")

```

5b) Optional extra credit: Moulins: Consider a 2.5 km deep, 1 m radius moulin in Greenland, and suppose all the heat due to the potential energy of the meltwater flowing down the moulin is used to widen the moulin. What water flow, in m³/sec, is required to enlarge the radius to 2 m during the one month of peak melt season?

[]: